

High-Level Synthesis (HLS) with Vivado

Objective:

Translate a C-like algorithm into a synthesizable Verilog design using HLS tools.

Project Overview:

Students will:

- 1) Given a hardware-friendly format.
- 2) Use an HLS tool (e.g., Vivado HLS, LegUp, or MyHDL) to generate Verilog.
- 3) Simulate and verify the Verilog design.
- 4) Collect performance metrics and explore the design space.

Given Project Files:

Project 1: Sobel Edge Detection Filter

Input : Grayscale image (e.g., 256x256 pixels).

Output : Edge-detected image using the Sobel operator.

Your assignments:

- 1) Synthesize the design with the defaults. View the synthesis results and answer the question
 - a. Estimated clock period:
Worst case latency:
Number of DSP48E used:
Number of FFs used:
Number of LUTs used:
 - b. Select **Solution > Open Analysis Perspective** or click on *Analysis* button on tools bar to open the analysis viewer.
 - c. Select the **Resource Profile** tab and observe various resources and where they have been used.
- 2) Run C/RTL Co-simulation.

- a. Once the simulation verification is completed, the simulation report tab will open showing the results. The report indicates if the simulation passed or failed. In addition, the report indicates the measured latency and interval. Since we have selected only VHDL, the result shows the latencies and interval (initiation) which indicates after how many clock cycles later the next input can be provided. Since the design is not pipelined, it will be latency+1 clock cycles. You should see something like the following:

Cosimulation Report for 'matrixmul'

Result

RTL	Status	Latency			Interval		
		min	avg	max	min	avg	max
VHDL	Pass	79	79	79	NA	NA	NA
Verilog	NA	NA	NA	NA	NA	NA	NA

Export the report(.html) using the [Export Wizard](#)

- b. Analyze the dumped traces. You should have something like the following:

